



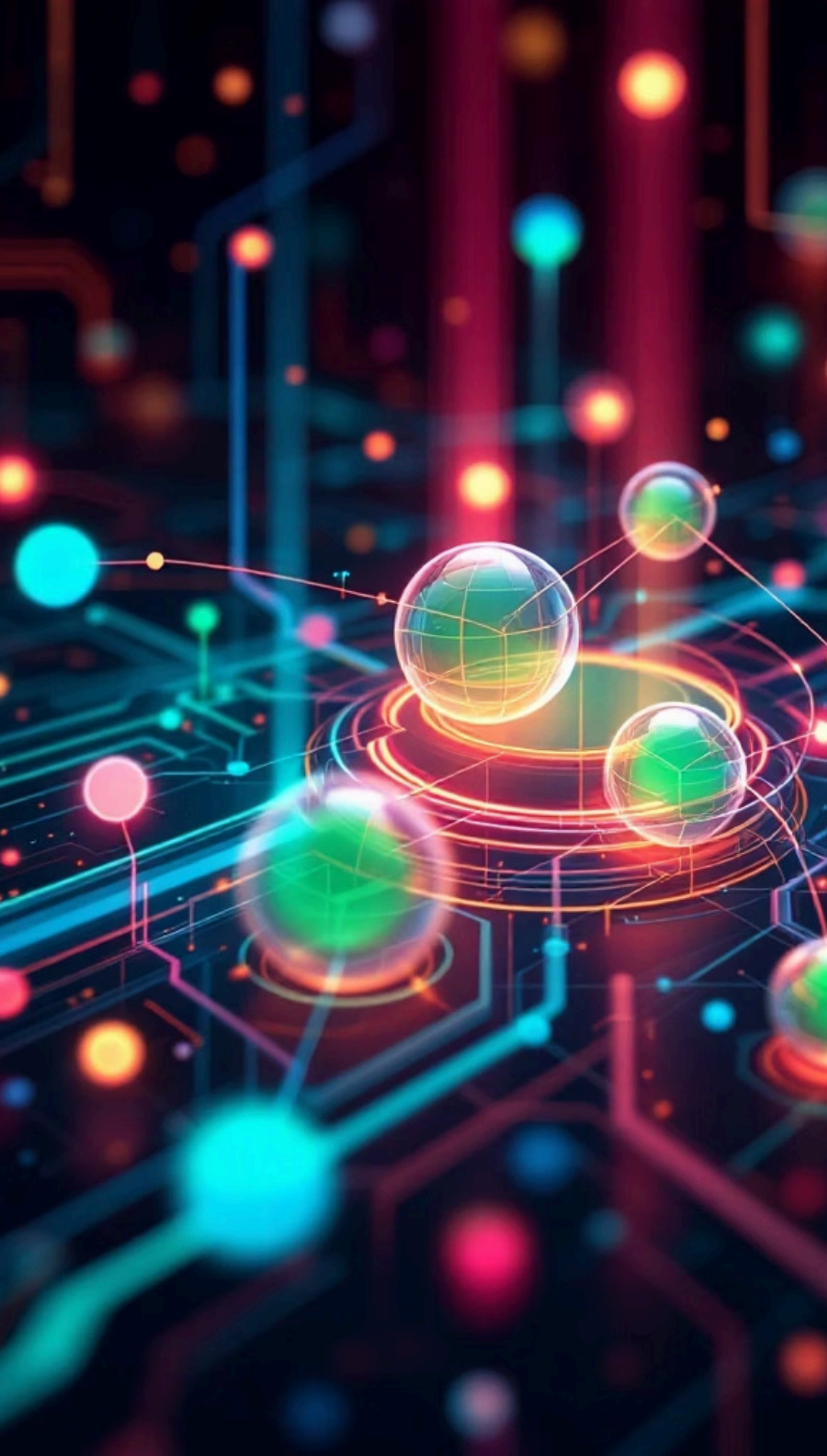
Personalized Education Agent

DATA 298B MSDA Project II – Team 2

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Model Development Overview

The project builds a smart academic support system using **eight open-weight LLMs** across five families — balancing response quality, reasoning, coding skill, and deployment efficiency.

3 Core Priorities

Educational responses, fine-tunability within academic compute limits, and selective multi-model deployment.

5 Model Families

Mistral, Llama, Qwen, Gemma, and DeepSeek — each serving distinct tutoring roles.

Role-Based Design

Smaller models for speed; larger models for reasoning depth and richer explanation.

Model Proposals: The Faculty of LLMs

Each model was selected as a module in a multi-model tutoring framework, evaluated on instructional quality, reasoning depth, coding support, efficiency, and adaptability.

Model	Size	Primary Role	Key Strength	Main Limitation
Mistral-7B-Instruct-v0.3	7B	Balanced baseline	Instruction quality + tuning feasibility	Less specialized than Qwen/DeepSeek
Llama-3.2-3B-Instruct	3B	Efficient concept support	Low resource, fast inference	Reduced reasoning depth
Llama-3.1-8B-Instruct	8B	Deep concept explanation	Strong reasoning flow	Higher latency and compute
Qwen 3.5 4B	4B	Efficient coding support	Technical capability + runtime efficiency	Less detailed than 9B
Qwen 3.5 9B	9B	Advanced coding pathway	Strongest coding assistance	Higher memory/runtime cost
Gemma-2-2B-it	2B	Quick clarifications	Very efficient, deployment-friendly	Limited depth on advanced tasks
Gemma-2-9B-it	9B	Richer tutoring output	Better detail than 2B variant	Slower, more resource-intensive
DeepSeek-R1 7B Distilled	7B	Advanced reasoning pathway	Strong logical decomposition	Not ideal for simple queries

Model Support Environment

Development Stack

GPU-enabled Python notebook environment with PyTorch, Hugging Face Transformers, PEFT, LoRA/QLoRA, and BitsAndBytes for quantized loading.



Figure 4.1. Data Preparation and Fine-Tuning Workflow

Fine-Tuning

Adapter-based PEFT – no full retraining. Modular, swappable adapters per tutoring role.

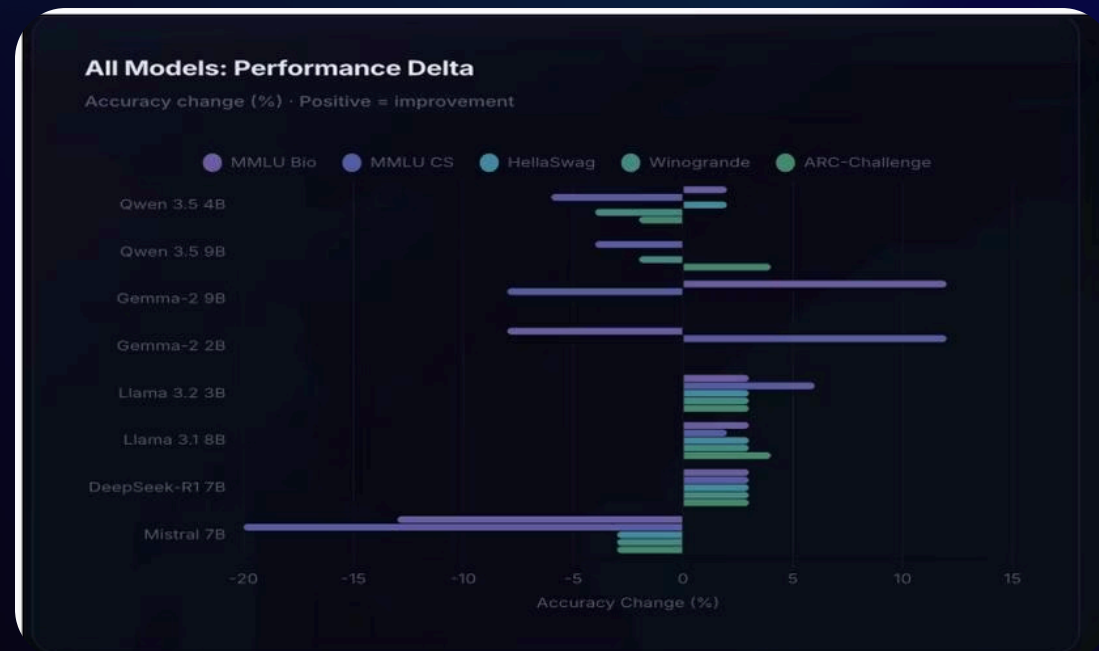
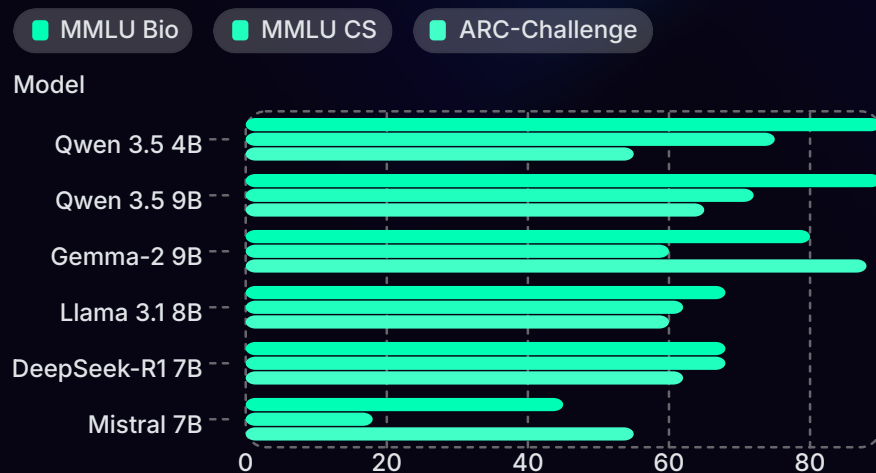
Quantization

Reduced-precision loading for 7B–9B models, keeping experimentation feasible.

Evaluation

MMLU, ARC-Challenge, HellaSwag, Winogrande, HumanEval, ROUGE-L, BLEU-4, BERTScore-F1.

Model Comparison & Benchmark Results

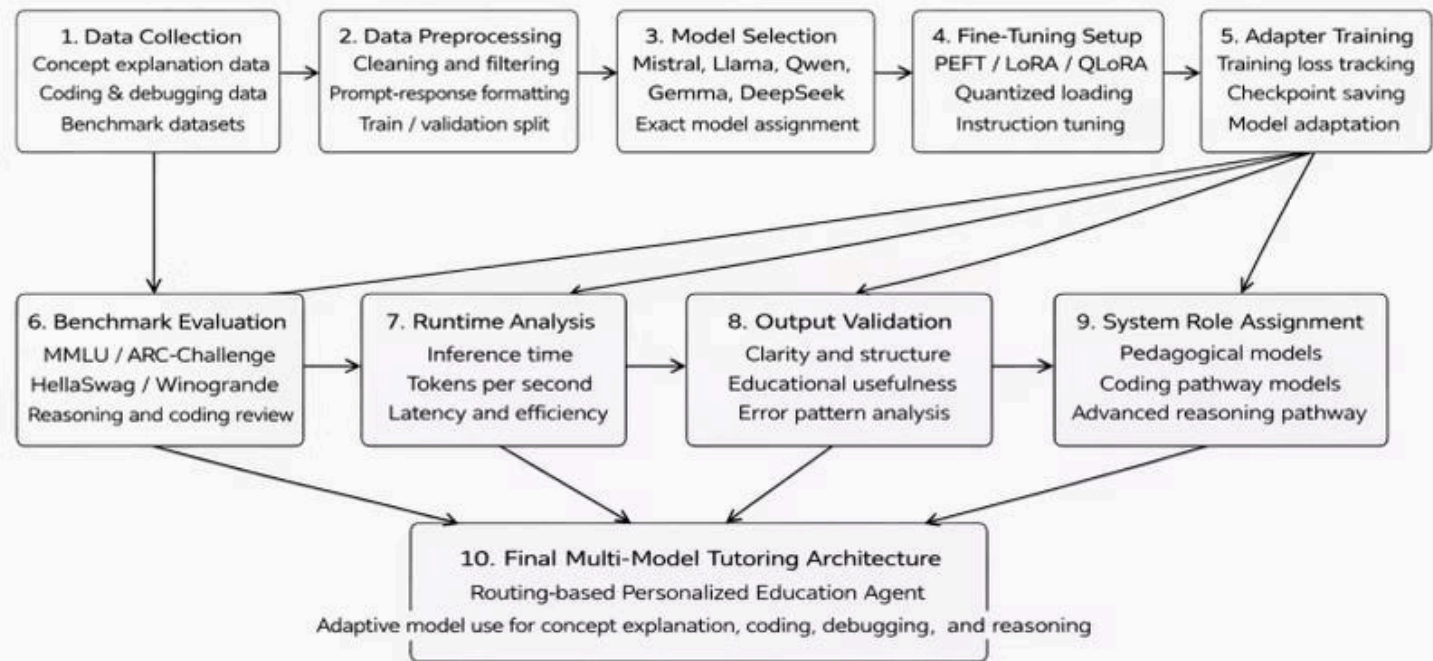


Larger models consistently outperform smaller variants on reasoning and knowledge tasks. **Gemma-2 9B leads ARC-Challenge (88%)**; Qwen 3.5 4B/9B dominate MMLU Bio (~90%). Mistral 7B shows the widest negative delta post fine-tuning.

Evaluation Framework

This multi-layer approach avoids reducing all model decisions to a single score – each model is assessed on its actual contribution to the tutoring system.

Figure 4.2. Model Development and Evaluation Pipeline



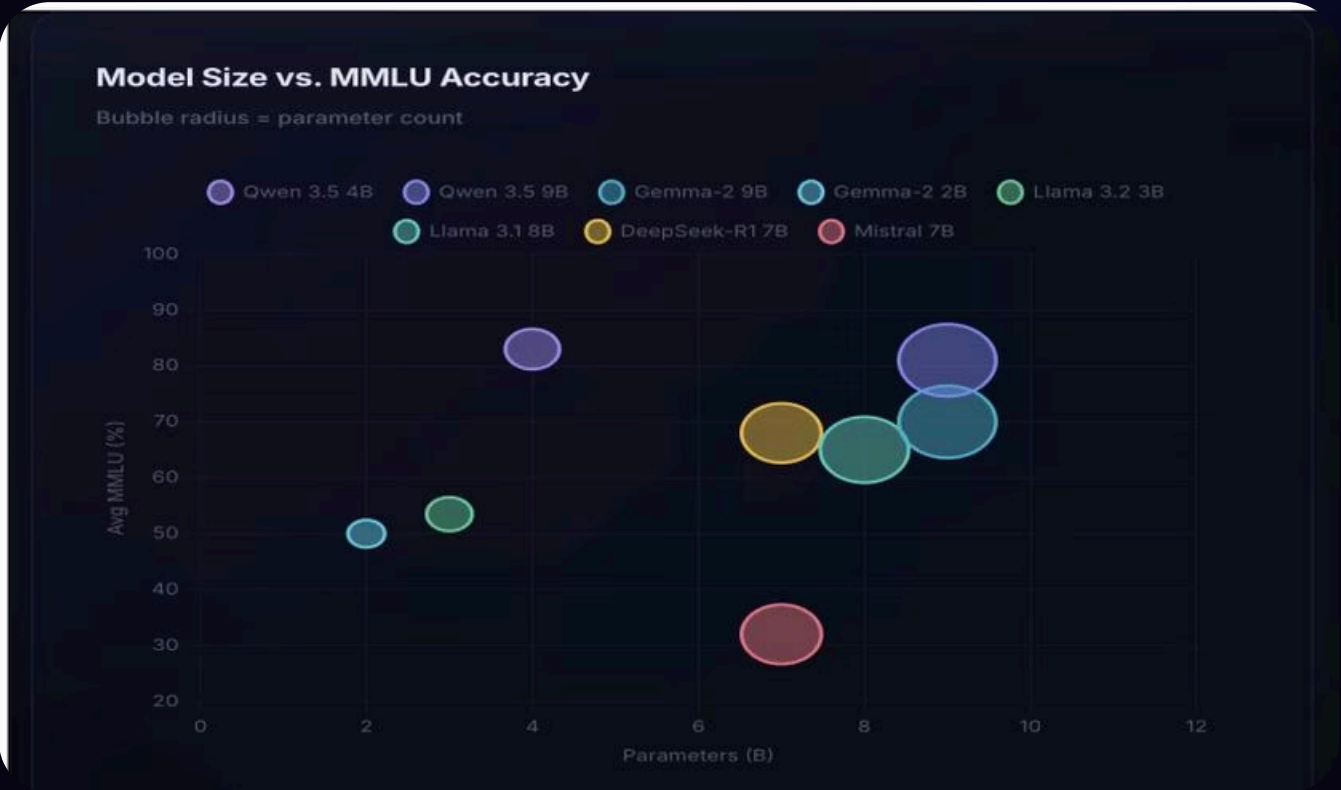
This pipeline shows how educational data is transformed into fine-tuned models, evaluated across quality and efficiency metrics, and integrated into the final tutoring system. This pipeline.

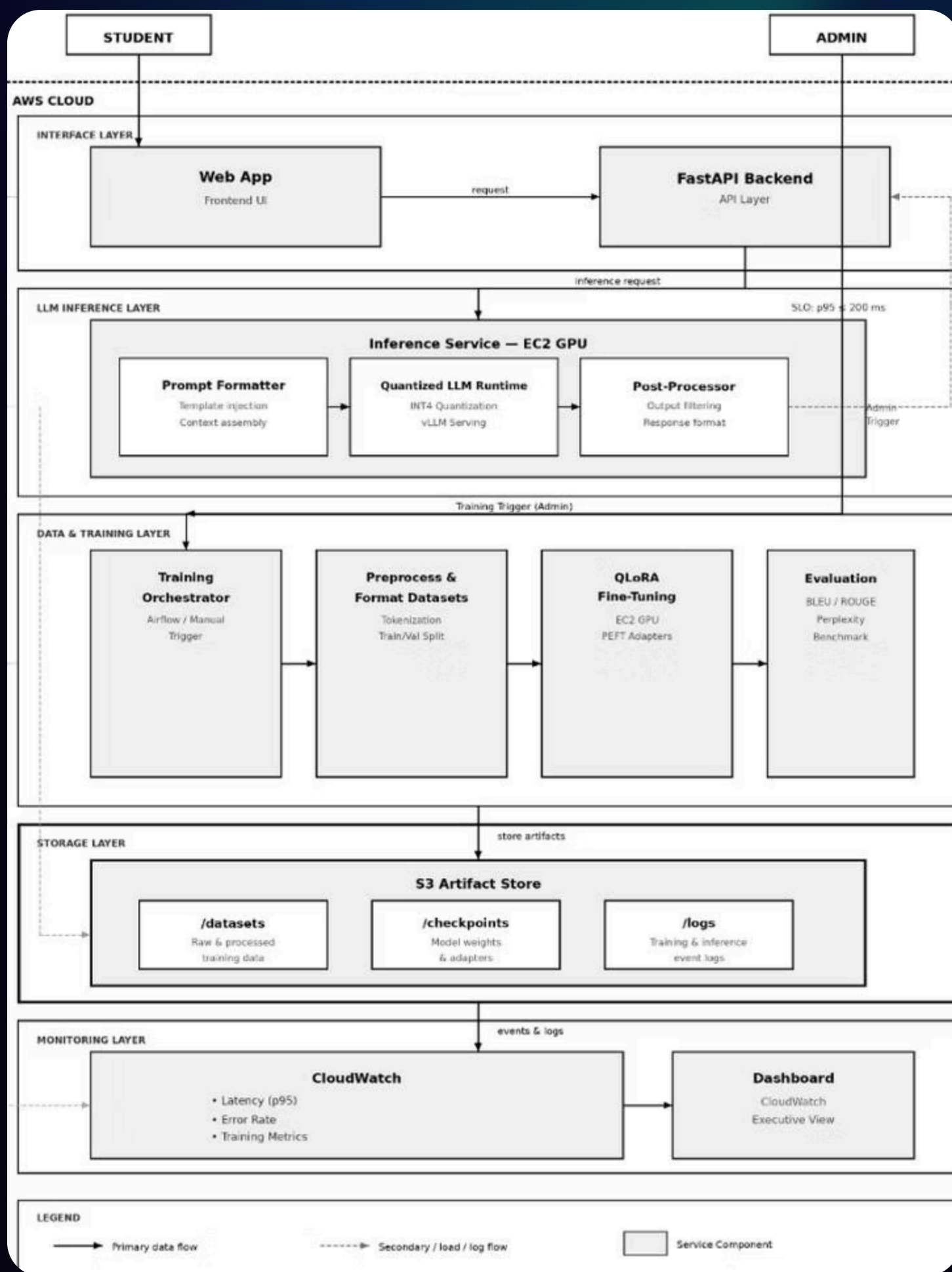
Validation Results: Size vs. Performance

Key Findings

- ➔ **Larger models** yield stronger reasoning, richer explanations, and better multi-step support.
- ➔ **Smaller models** (Gemma-2-2B, Llama-3.2-3B) remain highly useful for low-latency, lightweight interactions.
- ➔ **Qwen 3.5 4B** punches above its weight — ~83% avg MMLU at only 4B parameters.
- ➔ **Mistral 7B** shows the lowest MMLU avg (~32%), confirming it as a general baseline rather than a specialist.
- ➔ **Fine-tuning** improved tutoring alignment across all models without full retraining.

📌 Gemma-2 2B achieves **6.6 tokens/sec** — fastest inference in the set. Mistral 7B is slowest at 2.8 tokens/sec.





System Architecture: 5-Layer Design

User Interaction

Accepts natural language, code snippets, and debugging prompts.

Application Layer

Validates, tags, and routes requests; formats responses.

Routing & Orchestration

Identifies task type; selects optimal model pathway.

Model Inference

Runs selected model with tutoring-oriented structured prompts.

Storage & Monitoring

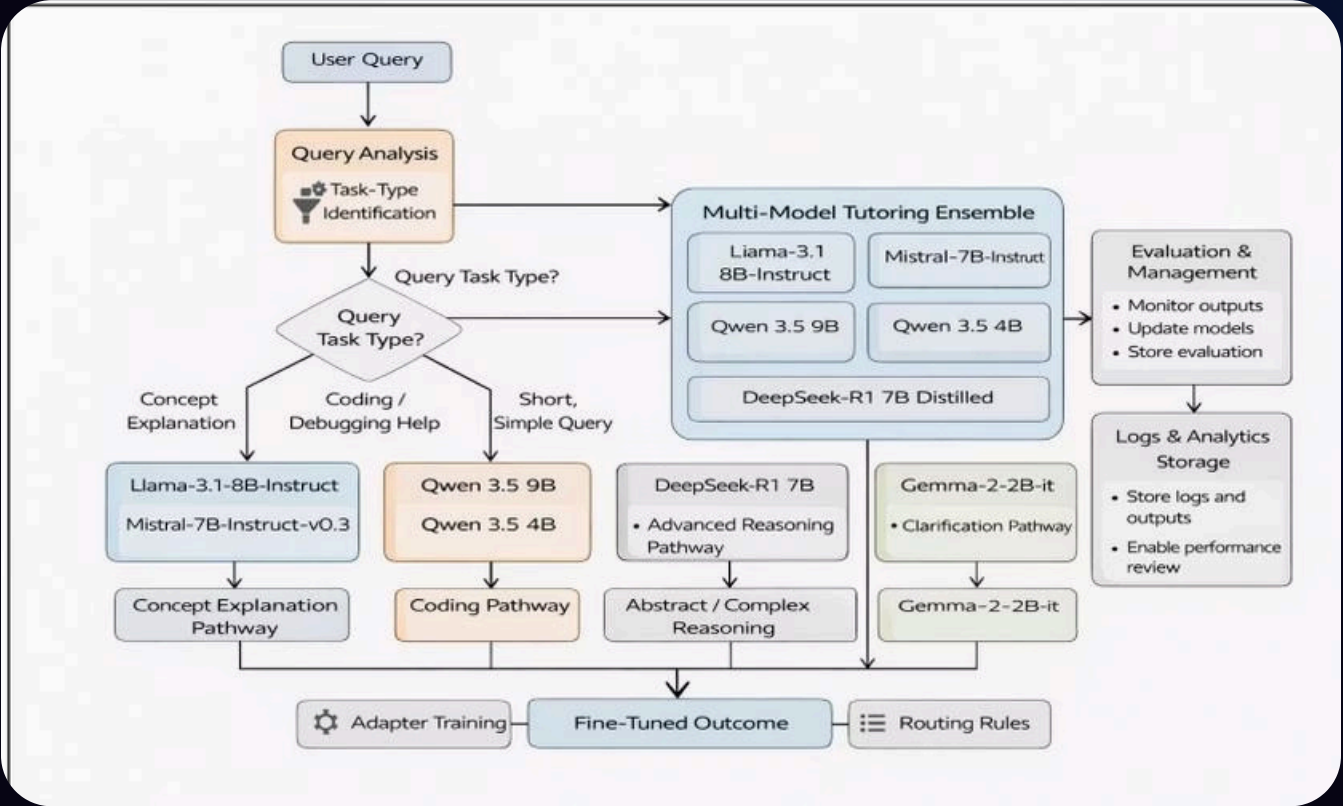
Logs queries, outputs, adapters, and performance metrics.

Intelligent Routing: The Core of Adaptive Tutoring

Model-to-Task Routing Strategy

Query Type	Assigned Model(s)
Short clarification	Gemma-2-2B-it, Llama-3.2-3B
General concept explanation	Mistral-7B, Llama-3.1-8B
Deep conceptual reasoning	Llama-3.1-8B, DeepSeek-R1 7B
Coding help	Qwen 3.5 4B, Qwen 3.5 9B
Debugging assistance	Qwen 3.5 9B, Mistral-7B
Advanced technical reasoning	DeepSeek-R1 7B Distilled

This routing logic converts model comparison findings into real adaptive tutoring behavior – matching computational cost to educational need.



Achievements, Constraints & Final Architecture

Major Achievements

- Rigorous multi-family LLM comparison across reasoning, coding, and pedagogy
- PEFT fine-tuning improved tutoring alignment without full retraining
- Multi-layer evaluation: benchmarks + runtime + qualitative review
- Layered, modular architecture ready for future personalization
- Role-based routing validated through functional system testing

Key Constraints

- Academic compute limits constrained tuning scope
- No large-scale real-user classroom deployment yet
- Full learner-aware personalization is future work
- Routing logic can be further refined with confidence-based classification

Conclusion

No single model is best for all tasks. The **multi-model, role-based architecture** is the most realistic and adaptive design for a Personalized Education Agent — matching each model to where it adds the most value.